

Focused Contrastive Loss for Classification With Pre-Trained Language Models

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Abstract—Contrastive learning, which learns data representations by contrasting similar and dissimilar instances, has achieved great success in various domains including natural language processing (NLP). Recently, it has been demonstrated that incorporating class labels into contrastive learning, i.e., supervised contrastive learning (SCL), can further enhance the quality of the learned data representations. Although several works have shown empirically that incorporating SCL into classification models leads to better performance, the mechanism of how SCL works for classification is less studied. In this paper, we first investigate how SCL facilitates the classifier learning, where we show that the contrastive region, i.e., the data instances involved in each contrasting operation, has a crucial link to the mechanism of SCL. We reveal that the vanilla SCL is suboptimal since its behavior can be altered by variances in class distributions. Based on this finding, we propose a Focused Contrastive Loss (FoCL) for classification. Compared with SCL, FoCL defines a finer contrastive region, focusing on the data instances surrounding decision boundaries. We conduct extensive experiments on three NLP tasks: text classification, named entity recognition, and relation extraction. Experimental results show consistent and significant improvements of FoCL over strong baselines on various benchmark datasets, especially in few-shot scenarios.

Index Terms—Data mining, text mining, natural language processing, and contrastive learning.

I. INTRODUCTION

CLASSIFICATION is a key task in machine learning, which aims to assign the correct class label for each data instance from the input. It has many applications in various domains, including natural language processing, computer vision, speech recognition, and recommender systems. However, constructing an effective classifier is non-trivial, as it requires to uncover

the complex correlations presented in the input. Over the past years, many attempts have been made to improve the performances of classification models by developing more effective model architectures, especially leveraging deep learning [1], [2]. Notably, in the domain of natural language processing, various pre-trained language models such as BERT [3] and RoBERTa [4] have gained great success for text-based classification tasks, such as text classification, named entity recognition, and relation classification. These language models are pre-trained on large scale unlabelled data using self-supervised learning, and can be transferred to downstream supervised tasks (e.g., classification tasks) via fine-tuning.

Although the use of pre-trained language models for classification tasks have gained remarkable success [5], [6], [7], the finetuning process is still challenged by the noise and variances in the training data of downstream tasks. In particular, real-life classification datasets are usually highly imbalanced with uneven number of data instances for different classes. As such, developing classification models that are robust against the class distribution variances is of great importance. Furthermore, the finetuning process usually requires considerable amount of labelled data instances to serve as the supervision signal in training, which are expensive to obtain, especially considering the domain expertise that is needed for data annotation and the infinite space of all possible classification tasks. Therefore, few-shot learning, which aims to let a machine learning model perform a new task with limited number of labelled data for that task, has attracted extensive attention recently [8], [9], [10], [11], [12].

To address these challenges, in this paper, we focus on the loss function design for classification, which is undoubtedly a crucial step in finetuning since the entire training process is guided by the loss function. For classification models, an ideal loss function is expected to guide the model to learn discriminative features of data, so that data instances from different classes can be easily separated.

Cross-entropy (CE) is the most popular classification loss function. As a derivative of maximum likelihood estimation, CE operates under the i.i.d. assumption of data. However, statistical heterogeneity is common in real-world datasets, e.g., high correlations and class imbalance among data, which invalidates this i.i.d. assumption. Moreover, CE is accuracy-oriented. That is, CE prioritises classifier performance rather than explicitly examining whether the features extracted from data are discriminative enough. We conduct experiments on a simulated

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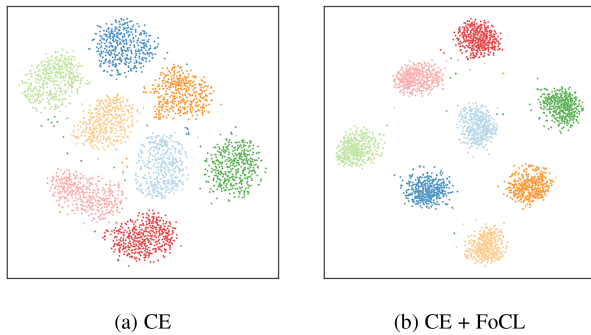


Fig. 1. t -SNE visualization of trained embeddings on a simulated classification dataset with 8 classes. Compared with CE, CE + FoCL (proposed) reduces inner-cluster variations and increases the inter-cluster distances, facilitating the learning of decision boundaries. The simulation setup is detailed in Section VIII.

dataset with 8 classes, and visualize the trained embeddings with CE in Fig. 1(a). We see that while data instances form separate clusters for different classes, these clusters are loose, with small inter-cluster margins. This indicates that the deep neural network is not optimized to learn the most discriminative features under CE. As such, it is difficult to build accurate classifiers, especially considering the negative instances near decision boundaries (*hard negatives*).

Various loss functions have been proposed to improve CE [13], [14]. A common approach is to weigh the training instances in relation to the overall loss, so that the model focuses more on the hard negative instances near the decision boundaries rather than those instances that are easy to classify. However, such a re-weighting strategy has two disadvantages. First, the weights for different classes/instances need to be pre-defined, involving one or more hyperparameters that need to be tuned. Second, these loss functions still do not explicitly examine the quality of features used for classification.

Recently, several works [15], [16], [17] have shown that incorporating supervised contrastive learning (SCL) [18] into NLP classification tasks can improve prediction accuracy via directly optimizing the distributions of data instances in the deep latent space. The purpose of using SCL is to pull instances from the same class close to each other in the latent space and push instances from different classes away from each other. Hence, it guides the model to map instances from different classes far away in the latent space. This explicitly makes the model to learn more separable features for different classes, facilitating the learning of decision boundaries.

Although good empirical results have been reported, the mechanism of SCL is less studied. In this paper, we first perform gradient analysis to examine the behavior of SCL, and how it optimizes a model for classification. We show that the contrastive region, i.e., the data instances that are involved in each contrasting operation, has a crucial link to the behavior of SCL. From the analysis, we reveal that the behavior of SCL will be altered when class distribution within the contrastive region changes. For example, when a sampled batch is biased towards one class, SCL may push the instances from the same

class away from each other in the latent space, instead of pulling them together, making it suboptimal for classification tasks.

Based on our findings, we propose a new contrastive loss for classification, named **focused contrastive loss (FoCL)**. Compared to SCL, FoCL defines a more focused contrastive region in the latent space, contrasting data instances near the decision boundaries. As such, FoCL is guaranteed to optimize cluster distributions of different classes in the latent space even when the class distribution changes, and thus, FoCL is more robust to variance in class distribution. As shown in Fig. 1(b), the trained clusters using CE + FoCL are much tighter than those with CE only. This demonstrates that FoCL guides the model to learn more discriminative features from data. A further comparison between SCL and FoCL is included in Section VIII.

The contributions of this paper are as follows:

- 1) We study the behavior of SCL for classification via analyzing its gradients. We demonstrate how the presence of class distribution variation alters SCL's behavior and deteriorates the model performance.
- 2) We propose a new contrastive loss function, FoCL, which defines a finer contrastive region focusing on the data instances near decision boundaries. We prove that FoCL improves over SCL in terms of its robustness against class distribution variation.
- 3) We validate our approaches on three classification tasks: text classification, named entity recognition (NER) and relation extraction (RE). We also experiment on the task of joint modelling of NER and RE. The experimental results show that FoCL outperforms strong baselines significantly, especially under few-shot scenarios. Our code is publicly available online at [GitHub_Link_Withheld](#).

II. RELATED WORK

This paper is closely related to several streams of works including classification based on pre-trained language models, contrastive learning, and loss functions for classifications. In this section, we review the three groups of related studies respectively.

A. Classification Using Pre-Trained Language Models

Classification tasks are common in the domain of natural language processing, including classification on the sentence level (e.g., sentence classification [19], [20], [21] and sentiment analysis [22], [23]), token level (e.g., part-of-speech tagging [24], [25] and named entity recognition [26], [27]), and level of pairwise relations (e.g., relation classification/extraction [28], [29]). Developing accurate classification models for these tasks is challenging, since the vocabulary size is large, and the semantic meanings of words/sentences are heavily dependent on the contexts. Therefore, pre-trained contextual word embeddings are popular choices for input encoders in the literature.

With the great success of the Transformer neural architecture [30], various architectures for pre-trained language models have been proposed, including BERT [3], RoBERTa [4], and GPT3 [31]. The main idea is to train a language model on unsupervised learning tasks (e.g., masked language modelling

and next sentence prediction) using the large volume of online corpus. From this, the language model can learn contextualized representations for words and sentences, which can be transferred to downstream tasks including classification tasks. When finetuning for a new task, the language model can be used as the input encoder, with a few additional task-specific modules to perform the new task.

It has also been shown that when a corpus from a specific domain is used in the pre-training, a language model that is more specialized in that text domain can be obtained, e.g., BERTweet [6] for twitter texts, SciBERT [32] for scientific literature, and BioBERT [33] for bio-clinical literature.

B. Contrastive Learning

Contrastive learning was originally proposed as an unsupervised learning method for images. The core idea is that if two data instances are similar (i.e., an image and a rotated version of the same image), their latent representations should be similar (i.e., located closely to each other in the latent space). Based on this intuition, contrastive learning aims to learn representations of data by contrasting similar and dissimilar data instances without explicit supervision labels [34], [35].

Inspired by the success of unsupervised contrastive learning, supervised contrastive learning (SCL), which incorporates with class labels, was proposed [18]. SCL considers data instances from the same class as similar instances, and instances from different classes as dissimilar instances. Hence, SCL will map data instances from the same class closely in the latent space while keeping instances from different classes far away. As such, SCL guides a model to cluster instances from the same class together, allowing the model to learn the separable features of different classes.

C. Loss Functions for Classification

The most common loss function for classification is the CE loss [36], which measures the difference between the predicted label distribution of one data instance and its ground-truth label distribution. However, studies have suggested that CE is sub-optimal for real-life classification datasets, especially when the data is not independent and identically distributed (e.g., when there exists high class imbalance) [14].

Many attempts have been made to improve CE in terms of its robustness against uneven class distributions. A common approach is to re-weight the importance of data instances in relation to the overall loss function, which can be seen as a variant of CE but with weight assigned to each instance representing the importance of that instance. Re-weighting approaches can be categorized into two groups: hard and soft re-weighting. The hard re-weighting aims to select instances that need to be involved in the training process. Some of the early attempts include using oversampling [37] to balance the class distributions, and hard sample mining [38] which identifies the most difficult samples to challenge the classifiers. For soft re-weighting strategy, all data instances will participate in the training process, however, they are assigned with different weights based on their importance to model training. Therefore, for this group

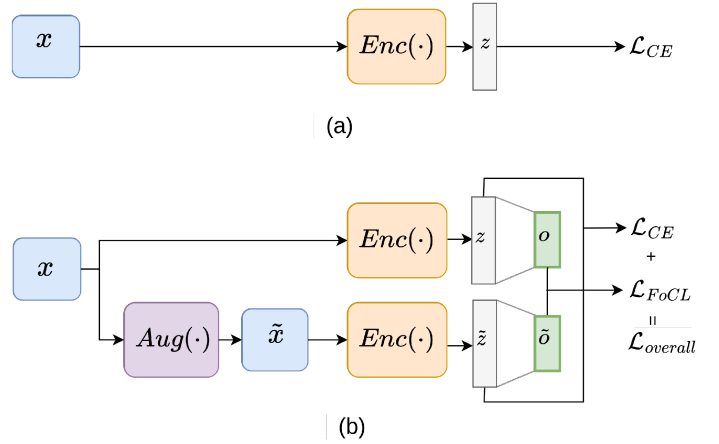


Fig. 2. (a) Scheme of a typical classification model. (b) Scheme of the proposed framework for FoCL. x : input instance; $\tilde{x}$: augmented version of x ; $Aug(\cdot)$: data augmentation module; $Enc(\cdot)$: (pre-trained) encoder.

of methods, it is crucial to identify a scheme that can be used to judge the importance of each data instance. Kumar et al. propose a self-paced learning scheme where weights of instances are obtained through optimizing the weighted training loss [39]. Lin et al. [14] propose a soft weighting scheme which penalizes the model more if the model misclassifies a data instance near the decision boundaries.

The aforementioned loss functions primarily focus on optimizing the classifier's accuracy, and does not explicitly consider the distribution of the latent representations of data instances. Recently, several studies have shown that directly optimizing the data distributions in the latent space can better facilitate the finetuning of pre-trained encoders for classification tasks [15], [16], [17]. Specifically, these studies combine CE with SCL, where CE focuses on optimizing the classifier, and SCL focuses on directly optimizing the underlying distributions of data instances in the latent space. Our work belongs to this latter group of study. However, in this paper, we reveal that the behavior of SCL has a crucial link to class distributions in the dataset. Therefore, directly incorporating SCL into classification tasks may yield suboptimal results.

III. PRELIMINARIES

A. Problem Setting

Let $\{x_i, y_i\}_{i=1, \dots, N}$ be a data sample, where x_i denotes an input data instance and y_i denotes the corresponding ground-truth label, and N denotes the total number of samples. Here, y_i represents a label belonging to a set of predefined classes: $y_i \in [C] = \{1, 2, \dots, C\}$. We aim to learn a scorer $f: \{x_i\}_{i=1, \dots, N} \rightarrow \{y_i\}_{i=1, \dots, N}$ that minimizes the *instance classification error*.

Fig. 2(a) illustrates a typical classification model based on a pre-trained language model. Given an input x , a pretrained encoder $Enc(\cdot)$ is used to embed x in a latent space. Let z be the embedding produced by $Enc(x)$. A classifier, e.g., a linear projection layer, is then used to map z to the predicted label $\hat{y}$. The choice of encoder here can be any pre-trained language model depending on the corpus domain of the classification task.

During training, the pre-trained encoder is fine-tuned together with the classifier via a *loss function*.

B. Cross-Entropy Loss

CE aims to maximize the probability of an instance x_i being classified as its ground-truth label y_i , which can be formulated as [36]

$$\mathcal{L}_{CE} = - \sum_i \log \Pr(x_i, y_i), \quad (1)$$

where $\Pr(x_i, y_i)$ represents the probability of classifying x_i as y_i . Although the CE loss is widely used, it has two weaknesses. First, CE assumes each training sample contributes equally to the loss function. Hence, it is not ideal for datasets with high class imbalance as it favors the over-represented class. Second, CE is accuracy-oriented and does not explicitly consider how the latent representations of instances are distributed in the latent space. Models trained with CE further do not consider the tightness of clusters directly and hence are vulnerable to noisy data near decision boundaries.

C. Re-Weighting Strategy

To improve CE on imbalanced dataset, various strategies have been proposed, including under-sampling the over-represented class based evolutionary algorithms [13] and boosting algorithms that weight samples based on the classification error types [40]. Recently, Lin et al. [14] propose the focal loss (FL) to weigh training samples based on the difficulty of classifying them

$$\mathcal{L}_{FL} = - \sum_i (1 - \Pr(x_i, y_i))^\gamma \log \Pr(x_i, y_i), \quad (2)$$

where γ is a hyperparameter that controls the intensity of the re-weighting factor. When $\gamma = 0$, FL is equivalent to CE, while setting $\gamma > 0$ reduces the relative loss of the data instances that are easy to classify.

D. Supervised Contrastive Loss

Contrastive learning (CL) was first proposed for unsupervised learning [34], [35]. It aims to learn effective latent representations of data instances via contrasting data instances. Given an anchor data instance x_i , CL pulls x_i close to its *positive instances*, e.g., its augmented versions, while pushing it away from its *negative instances*, e.g., instances that are drastically different from x_i .

Khosla et al. [18] proposed later a supervised classification loss function that relies on CL, namely supervised contrastive loss (SCL). Given an anchor instance x_i , SCL assumes that all instances within the same class of x_i are positive instances w.r.t. x_i , and all instances from other classes are negative instances w.r.t. x_i . SCL is formulated as

$$\mathcal{L}_{SCL} = - \sum_i \frac{1}{|P_i|} \sum_{x_p \in P_i} \log \frac{\exp(s_{i,p}/\tau)}{\sum_{x_a \in A_i} \exp(s_{i,a}/\tau)}. \quad (3)$$

TABLE I
FREQUENTLY USED SYMBOLS

Symbol	Description
x_i	an anchor instance
P_i	positive instances of x_i (instances from the same class of x_i)
N_i	negative instances of x_i (instances from a different class than x_i)
A_i	all instances except for x_i ($P_i \cup N_i$)
$s_{i,j}$	similarity between the L_2 -normalized representations of x_i and x_j
x_p	a positive instance from P_i
x_n	a negative instance from N_i
I	total number of instances in the training set

Here, $s_{i,j} = \mathbf{z}_i^\top \mathbf{z}_j$ represents similarity between the L_2 -normalized representations $\mathbf{z}_i$ and $\mathbf{z}_j$ of the instances x_i and x_j . P_i represents the set of positive instances of x_i , A_i is the set of all instances excluding x_i , and $\tau \in \mathbb{R}^+$ is a temperature hyperparameter. Note that we use the operator $|\cdot|$ to represent the cardinality of a set.

Several existing works [15], [16], [17] demonstrate that incorporating SCL into classification tasks can improve classification accuracy via the joint use of CE and SCL

$$\mathcal{L}_{overall} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{SCL}, \quad (4)$$

where $\lambda \in (0, 1)$ is a hyperparameter controlling the contribution of SCL to the overall classification loss function. These works show that compared with CE, SCL can better optimize the distributions of instances in the latent space via capturing the similarity among instances from the same class and contrasting them with instances from other classes.

IV. UNDERSTANDING SCL FOR CLASSIFICATION

In this section, we investigate the behavior of SCL through analyzing its gradients. In particular, we examine the relationship between the class distribution and the behavior of SCL. We summarize the frequently used notations in Table I.

A. Gradients Derivation

For simplicity, we denote the probability of x_i being classified as x_j among all instances in A_i as

$$\Pr(x_i, x_j | A_i) = \frac{\exp(s_{i,j}/\tau)}{\sum_{x_a \in A_i} \exp(s_{i,a}/\tau)}. \quad (5)$$

Given *one anchor instance* x_i , the SCL loss for x_i can be formulated as

$$\mathcal{L}_{SCL}^i = - \frac{1}{|P_i|} \sum_{x_p \in P_i} \log \frac{\exp(s_{i,p}/\tau)}{\sum_{x_a \in A_i} \exp(s_{i,a}/\tau)}. \quad (6)$$

Gradient w.r.t. Negatives: Let $N_i = A_i \setminus P_i$ be the set of negative instances w.r.t. x_i , i.e., instances from different classes than that of x_i . Suppose x_n is one negative instance from N_i . The gradient of $\mathcal{L}_{SCL}^i$ w.r.t. the similarity $s_{i,n}$ between x_i and

x_n can be derived as

$$\frac{\partial \mathcal{L}_{SCL}^i}{\partial s_{i,n}} = \frac{1}{\tau} \Pr(x_i, x_n | A_i). \quad (7)$$

The process of obtaining the above gradient is detailed in Supplementary Section 2. Since $\Pr(x_i, x_n | A_i) \in (0, 1)$, the gradient in (7) is guaranteed to be *positive*. Thus, during training, the similarity $s_{i,n}$ will be minimized in order to minimize the overall SCL loss. This means SCL will push x_i away from its negative instance x_n . In addition, the magnitude of the above gradient is proportional to the probability of x_i being classified as x_n given A_i , i.e., $\Pr(x_i, x_n | A_i)$. This means that if the negative instance x_n is closer to x_i among all instances in A_i , the strength of such pushing will be larger.

Gradient w.r.t. Positives: Given a positive instance x_p that belongs to the same class of x_i , the gradient of $\mathcal{L}_{SCL}^i$ w.r.t. the similarity $s_{i,p}$ between x_i and x_p can be derived as

$$\begin{aligned} \frac{\partial \mathcal{L}_{SCL}^i}{\partial s_{i,p}} = & -\frac{1}{\tau |P_i|} \sum_{x_j \in N_i} \Pr(x_i, x_j | A_i) \\ & + \frac{1}{\tau |P_i|} \sum_{x_k \in P_i \setminus \{x_p\}} \Pr(x_i, x_p | A_i) - \Pr(x_i, x_k | A_i). \end{aligned} \quad (8)$$

The process of obtaining the above gradient is detailed in Supplementary Section 3. When the number of positives is small, i.e., $|P_i|$ being small, the second term (on the second line) in (8) is negligible. Therefore, the gradient is guaranteed to be *negative* since $\Pr(x_i, x_j | A_i) \in (0, 1)$. As such, during training, the similarity $s_{i,p}$ between anchor instance x_i and positive instance x_p will be maximized in order to minimize the loss, i.e., x_i will be pulled close to its positive instance x_p .

Furthermore, the magnitude of the first term is proportional to the summation of the gradients of x_i w.r.t. all negative instances in N_i . Hence, the strength of pulling will be larger if many negative instances are closer to x_i .

In summary, we see that in order for SCL to optimize the data distribution for classification tasks, the gradients of the anchor instance w.r.t. *negative instances* need to stay *positive*, while the gradients w.r.t. *positive instances* need to stay *negative*.

B. SCL Under Class Imbalance

In this section, we study the behavior of SCL when there is an unequal distribution of classes. The class distributions of real-world datasets are usually biased. Take NER datasets as an example. The majority of tokens are Outside of any named entities and hence, the ‘‘O’’-class is usually over-represented. That is, a given token x_i in the ‘‘O’’-class may have many positive instances and only a few (or no) negative instances in a sampled batch during training. Such variance in class distributions can negatively affect the behavior of SCL during training, as shown below.

Equation (8) shows that the gradient of anchor instance x_i w.r.t. a positive instance x_p consists of two terms. The first term is the summation of the gradients w.r.t. all negative instances, which is guaranteed to be negative as discussed in Section IV-A.

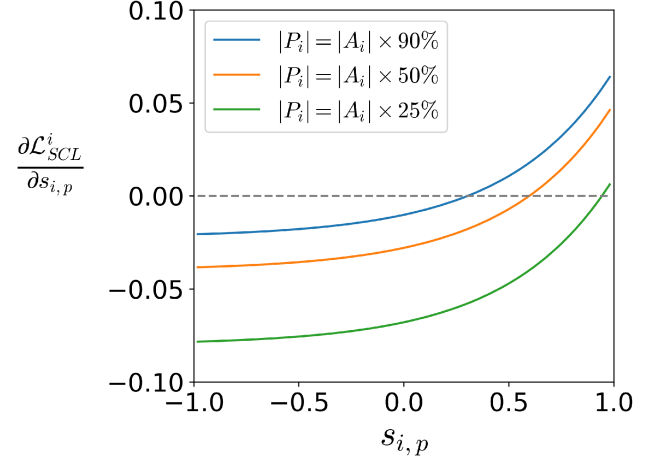


Fig. 3. Illustration of gradients w.r.t. positive instances with different compositions of A_i under SCL. When A_i contains more positive instances (e.g., $|P_i| = |A_i| \times 90\%$), the gradient w.r.t. a positive instance that is close to x_i (e.g., $s_{i,p} \rightarrow 1$) becomes positive, which means SCL will push x_i away from such a positive instance even though they are from the same class.

The second term represents the relationships between x_i and its positive instances. When the number of positive instances is small, the second term is negligible, e.g., the second term equals 0 if $|P_i| = 1$.

However, if there exist multiple positive instances in the sampled batch, the second term may become positive. Consider an extreme case: the positive instance x_p is close to the anchor instance x_i while all other instances in P_i and N_i are far from x_i in the latent space. Then, we have $\Pr(x_i, x_p | A_i) \gg \Pr(x_i, x_k | A_i), \forall x_k \in P_i \setminus \{x_p\}$ and $\Pr(x_i, x_p | A_i) \gg \Pr(x_i, x_j | A_i), \forall x_j \in N_i$. Hence, (8) can be simplified as follows:

$$\frac{\partial \mathcal{L}_{SCL}^i}{\partial s_{i,p}} = \frac{1}{\tau |P_i|} \sum_{x_k \in P_i \setminus \{x_p\}} \Pr(x_i, x_p | A_i). \quad (9)$$

Since $\Pr(x_i, x_p | A_i) \in (0, 1)$, the above gradient will become positive. Furthermore, the gradient magnitude will be larger if x_p is closer to x_i . This means that SCL will push this positive instance x_p away from x_i , instead of pulling them together.

We perform a simulation on the gradient w.r.t. positives in Fig. 3 to illustrate our finding. We sample one anchor instance as x_i and 100 data instances as $|A_i|$ in the latent space. The set A_i is uniformly sampled from the latent space. Then we consider three scenarios, where 90%, 50%, 25% of the instances in A_i are positive w.r.t. x_i .

We show the gradients of x_i w.r.t. all positives in Fig. 3 under SCL. We see that for all three cases, the value of the gradient of x_i w.r.t. a positive instance increases if the corresponding positive instance is closer to x_i (i.e., if $s_{i,p}$ is larger). Moreover, when there are more positives in A_i , the value of gradient increases more rapidly with $s_{i,p}$. In particular, when more than half of A_i is constituted by positive instances, the gradients w.r.t. many instances near x_i no longer stay negative. The positive gradient values suggest that those corresponding positive instances will

be pushed away from the anchor instance x_i , although they have the same class label as x_i .

C. Focused Contrastive Loss (FoCL)

To address this issue, we propose the focused contrastive loss (FoCL), improving the robustness of SCL to data imbalance with two strategies: 1) positive masking; and 2) hard negative mining.

1) *Positive Masking (PM)*: The first step is a simple yet effective way to improve the contrastive power within the contrastive region. We use an instance-wise mask to truncate the gradients w.r.t. redundant positive instances in (3). Given an instance x_i and one of its positive instances x_p , we force the gradients of all other positive instances, i.e., $P_i \setminus \{x_p\}$, to be zero and only keep the gradients of its negative instances. The resulting contrastive loss is

$$\mathcal{L}_{pm}^i = -\frac{1}{|P_i|} \sum_{x_p \in P_i} \log \frac{\exp(s_{i,p}/\tau)}{\exp(s_{i,p}/\tau) + E_-}, \quad (10)$$

where $E_- = \sum_{x_j \in N_i} \exp(s_{i,j}/\tau)$.

Gradient of PM w.r.t. Negatives: With positive masking, the gradient of the anchor instance x_i w.r.t. a negative instance x_n can be derived as

$$\frac{\partial \mathcal{L}_{pm}^i}{\partial s_{i,n}} = \frac{1}{\tau |P_i|} \sum_{x_p \in P_i} \Pr(x_i, x_n | \{x_p\} \cup N_i), \quad (11)$$

where:

$$\Pr(x_i, x_n | \{x_p\} \cup N_i) = \frac{\exp(s_{i,n}/\tau)}{\exp(s_{i,p}/\tau) + E_-}. \quad (12)$$

Since $\Pr(x_i, x_n | \{x_p\} \cup N_i) \in (0, 1)$, the gradient of x_i w.r.t. negative instance x_n is guaranteed to be positive. This means x_i is guaranteed to be pushed away from its negative instance x_n . Moreover, since for any $x_p \in P_i$, we have

$$\Pr(x_i, x_n | \{x_p\} \cup N_i) > \Pr(x_i, x_n | A_i). \quad (13)$$

Thus, the gradient magnitude in (11) is larger than that in (7). This means the strength of pushing x_i away from negative instances will be larger under PM than under the original SCL loss.

Gradient of PM w.r.t. Positives: The gradient of x_i w.r.t. a positive instance x_p under PM can be formulated as

$$\frac{\partial \mathcal{L}_{pm}^i}{\partial s_{i,p}} = -\frac{1}{\tau |P_i|} (1 - \Pr(x_i, x_p | \{x_p\} \cup N_i)). \quad (14)$$

The above equation shows that the gradient of the anchor instance x_i w.r.t. any positive instance x_p is guaranteed to be negative. This shows that with positive masking, x_i will be consistently pulled together to its positive instances throughout training.

2) *Hard Negative Mining*: The quality and quantity of negative instances are crucial to the performance of CL. Hard negative mining has been studied to identify high-quality negative instances [41], [42] in unsupervised settings. Inspired by these works, we propose to improve the term E_- in (10) via adding pseudo negative instances that are generated via hard negative mixing.

Specifically, from the set N_i of negative instances, we select a subset $H_i \subseteq N_i$ that consists of the hardest negatives, i.e., the top- K negative instances that are closest to x_i . That is, for any instance $x_h \in H_i$, the similarity of x_h and x_i , $s_{i,h}$, is no less than the similarity of x_i with any instances in $N_i \setminus H_i$. Here, K is a hyperparameter that controls the number of hard negative instances to be selected in our model.

With the hard negative set H_i , we generate M synthetic negative instances by mixing the embedding of anchor instance x_i with the hard negatives in H_i . Since all instances in H_i are from different classes than x_i , it is guaranteed that these synthesized negative instances will be located near decision boundaries.

We denote the set of synthesized hard negatives as $\tilde{H}_i$. Given a selected hard negative instance x_h from H_i , we generate one synthesized negative instance x_m via mixing the embedding $\mathbf{o}_i$ of the anchor instance x_i and the embedding $\mathbf{o}_h$ of the real hard negative instance x_h

$$\mathbf{o}_m = \frac{\alpha \cdot \mathbf{o}_i + (1 - \alpha) \cdot \mathbf{o}_h}{\|\alpha \cdot \mathbf{o}_i + (1 - \alpha) \cdot \mathbf{o}_h\|_2}, \quad (15)$$

where α is a random ratio. Note that we keep the value of α within $(0, 0.4)$ to ensure that the contribution of the anchor instance x_i itself is smaller than that of the hard negative instance. This means the generated pseudo points should be closer to the real negatives other than the anchor instance, since these pseudo points are used as negative instances.

Combining the positive masking and hard negative mining, the resulting FoCL w.r.t. to instance x_i is formulated as

$$\mathcal{L}_{FoCL}^i = -\frac{1}{|P_i|} \sum_{x_p \in P_i} \log \frac{\exp(s_{i,p}/\tau)}{\exp(s_{i,p}/\tau) + E_- + E_m}, \quad (16)$$

where $E_m = \sum_{x_m \in \tilde{H}_i} \exp(s_{i,m}/\tau)$ represents the contribution of the synthesized hard negatives. The overall loss is thus formulated as

$$\mathcal{L}_{overall} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{FoCL}, \quad (17)$$

where $\lambda \in (0, 1)$ is a hyperparameter controlling the weight of FoCL.

Gradient of FoCL w.r.t. Positives: Under FoCL, the gradient of x_i w.r.t. a positive instance can be formulated as

$$\frac{\partial \mathcal{L}_{FoCL}^i}{\partial s_{i,p}} = -\frac{1}{\tau |P_i|} (1 - \Pr(x_i, x_p | \{x_p\} \cup N_i \cup \tilde{H}_i)). \quad (18)$$

Since $\Pr(x_i, x_p | \{x_p\} \cup N_i \cup \tilde{H}_i) \in (0, 1)$, the above gradient is guaranteed to be negative.

Furthermore, we have

$$\Pr(x_i, x_p | \{x_p\} \cup N_i) = \frac{\exp(s_{i,p}/\tau)}{\exp(s_{i,p}/\tau) + E_-}, \quad (19)$$

and

$$\Pr(x_i, x_p | \{x_p\} \cup N_i \cup \tilde{H}_i) = \frac{\exp(s_{i,p}/\tau)}{\exp(s_{i,p}/\tau) + E_- + E_m}. \quad (20)$$

Hence, it is guaranteed that $\Pr(x_i, x_p | \{x_p\} \cup N_i) > \Pr(x_i, x_p | \{x_p\} \cup N_i \cup \tilde{H}_i)$. This shows that the gradient magnitude in

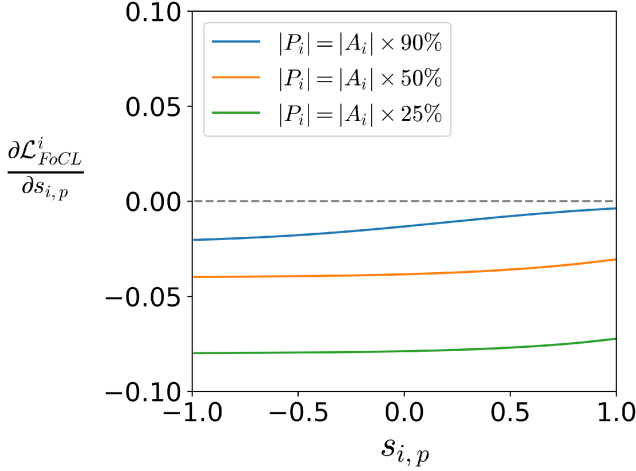


Fig. 4. Illustration of gradients w.r.t. positive instances with different compositions of A_i under FoCL. The gradients w.r.t. any positive instance stay negative, which means FoCL will always pull the anchor instance x_i to its positives.

(18) is larger than that in (14), which means that the strength of pulling the anchor instance closer to its positive instance will be larger for FoCL than PM.

Continuing with the simulation setup described in Fig. 3, we show in Fig. 4 the gradients w.r.t. positives of FoCL when 90%, 50%, and 25% of instances in $|A_i|$ are positive instances w.r.t. x_i . We see that under FoCL, the gradients of x_i w.r.t. its positives stay negative with different percentages of positive instances. This confirms our gradient analysis: FoCL is more robust against variances in class distributions, with consistent behavior of optimizing data distributions for classification tasks.

Contrastive Region of FoCL: From the above analysis, we see that given an anchor instance, the data instances that are involved in the computation of contrastive loss are critical to the performance of the loss function. We call the data instances involved in the loss computation as the *contrastive region* for the anchor instance. We illustrate the difference in contrastive regions of vanilla SCL and FoCL in Fig. 5, where the star represents the anchor instance x_i , crosses represent positive instances of x_i (instances with the same class label as x_i), circles represent negative instances of x_i (instances from a different class), and triangles represent pseudo negative instances. The contrastive region of SCL includes all positive and negative instances. On the other hand, the contrastive region of FoCL is squeezed from the grey rectangle to the grey ellipse. Moreover, the introduction of pseudo negative instances (red triangles) enhances the contrastive power within the new contrastive region and increases the margin between the two classes. We see that the contrastive region now is more focused on the decision boundary.

We further illustrate the working mechanism of FoCL in Fig. 6. In the figure, we keep only one positive instance (blue cross) and two negative instances (green circles) for ease of presentation. Since the gradient of FoCL w.r.t. positive instance is guaranteed to be negative, the anchor instance (star) will be pulled closer to the positive instance. On the contrary, the anchor instances will be pushed away from the negative instances. Furthermore, FoCL will generate pseudo negative samples by

mixing the anchor instance and real negative samples (red triangles), which are closer to decision boundary. During gradient update, FoCL will push the anchor instance further away from these pseudo negative samples. Thus, after the gradient update, the margin between the two classes will be increased, leading to tighter clusters.

V. GRADIENT COMPARISON OF SCL AND FoCL

We illustrate the derived gradients of SCL, PM, and FoCL in Table II. The key differences among the three loss functions can be summarized as follows.

Gradients w.r.t. Negatives: Given an anchor instance x_i , all three loss functions are guaranteed to have positive gradients for negative instances. This is because the term $\Pr(x_i, x_j | A_i)$, $\Pr(x_i, x_n | \{x_p\} \cup N_i)$, and $\Pr(x_i, x_n | \{x_p\} \cup N_i \cup \tilde{H}_i)$ are within the range of (0, 1). This will guarantee that all three loss functions will push x_i away from negative instances.

However, the gradient magnitude of PM is guaranteed to be larger than SCL. We provide the proof of this finding in Supplementary (Section 5). This means PM will be stronger in terms of pulling x_i away from its negative instances. Furthermore, FoCL enhances PM by introducing synthesized hard negatives, putting more focus on the most hard negatives that are closest to the anchor instance. Hence, the gradient magnitude of FoCL w.r.t. these most hard negatives will be further enhanced.

Gradients w.r.t. Positives: In Section IV-B, we have shown that the gradient of SCL w.r.t. positive instances will go above 0 when there are multiple positives. However, the gradients of PM and FoCL w.r.t. positive instances are guaranteed to be negative (see proof in Section 5 of Supplementary).

Finally, as discussed in Section IV-C2, the gradient magnitude of FoCL w.r.t. positive instances is higher than that of PM, which means FoCL will be stronger than PM in terms of pulling an anchor instance close to its positive instances.

VI. THE FRAMEWORK

To compute FoCL or SCL, each instance in a sampled mini-batch needs to have at least one positive instance (an instance with the same label). Some previous works satisfy this requirement via sampling [16], i.e., requiring the next instance to have the same/different label with existing instances in the mini-batch. However, this is not always feasible in NLP tasks, e.g., NER and the relation extraction (RE) task. In these tasks, sampling is conducted at sentence level but classification is performed at token/token pair level. Therefore, we use data augmentation to satisfy this requirement. For each sentence x in a randomly sampled mini-batch, we generate an augmented version $\tilde{x}$. This guarantees that each data instance to be classified will then have at least one positive instance in the augmented batch.

The overall framework is depicted in Fig. 2(b). Given an input x , an augmented version $\tilde{x}$ is generated. Then instance x and $\tilde{x}$ are forwarded to the same encoder, producing two embeddings z and $\tilde{z}$. These are then used in the computation of loss functions.

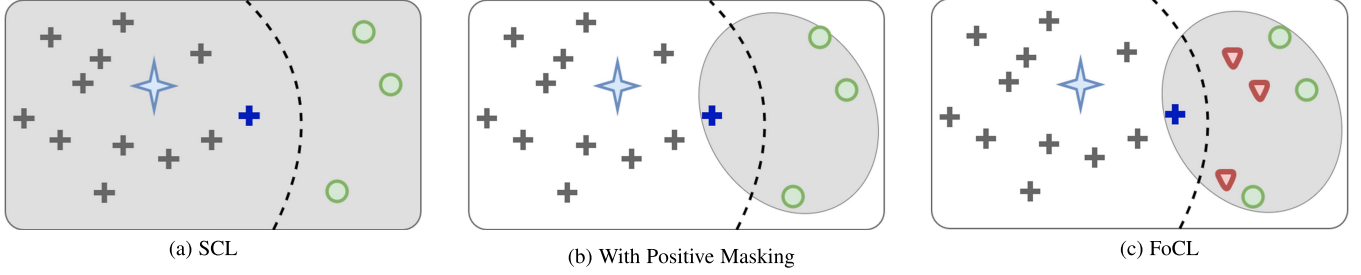


Fig. 5. Contrastive region of SCL is the entire grey rectangle enclosing all data instances (a). Contrastive region of FoCL, on the contrary, is the grey ellipse in (b)/(c), with a specific focus around the decision boundaries. Grey rectangle/ellipse: contrastive regions. Blue star: anchor instance x_i ; Crosses: positive instances P_i ; Blue cross: one positive instance x_+ ; Green circles: real negative instances N_i ; Red triangle: pseudo negative instances $\tilde{H}_i$.

TABLE II
SUMMARY OF GRADIENTS OF SCL, PM, AND FoCL

Loss	Gradient w.r.t. negatives ($\frac{\partial \mathcal{L}^i}{\partial s_{i,n}}$)	Gradient w.r.t. positives ($\frac{\partial \mathcal{L}^i}{\partial s_{i,p}}$)
SCL	$\frac{1}{\tau} \Pr(x_i, x_n A_i)$	$-\frac{1}{\tau P_i } \sum_{x_j \in N_i} \Pr(x_i, x_j A_i) + \frac{1}{\tau P_i } \sum_{x_k \in P_i \setminus \{x_p\}} \Pr(x_i, x_p A_i) - \Pr(x_i, x_k A_i)$
PM	$\frac{1}{\tau P_i } \sum_{x_p \in P_i} \Pr(x_i, x_n \{x_p\} \cup N_i)$	$-\frac{1}{\tau P_i } (1 - \Pr(x_i, x_p \{x_p\} \cup N_i))$
FoCL	$\frac{1}{\tau P_i } \sum_{x_p \in P_i} \Pr(x_i, x_n \{x_p\} \cup N_i \cup \tilde{H}_i)$	$-\frac{1}{\tau P_i } (1 - \Pr(x_i, x_p \{x_p\} \cup N_i \cup \tilde{H}_i))$

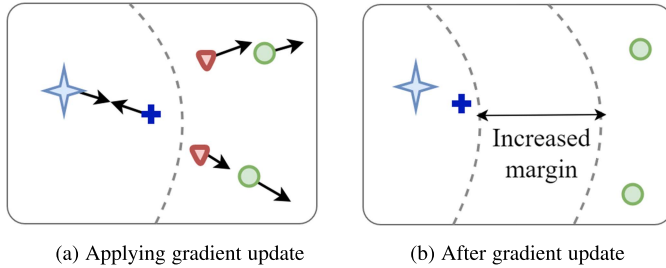


Fig. 6. Illustration of the working mechanism of FoCL.

To best test the effectiveness of proposed FoCL loss, we consider in this paper the simplest data augmentation method: noise induction by standard dropout [43]. Specifically, this augmentation method forwards a data instance x through a dropout layer twice, each time with *different* dropout masks.

VII. COST ANALYSIS

Let B be the number of samples in a mini-batch, C the number of classes, K the number of pseudo negative samples that are generated in hard negative mining (see Section IV-C2) of FoCL. The complexity of CE is $O(B \cdot C)$ since it needs to compute the dot-product between each sample with each class embedding.

FoCL needs to compute the dot-product between each sample with all other samples within the same mini-batch plus K pseudo negatives, which takes $O(B^2 + K \cdot B)$. The operation to find the top- K hardest negatives is bounded by $B \log B + K$. Hence, the overall complexity of FoCL is $O(B^2 + K \cdot B + B \log B + K)$, which demonstrates that the complexity of FoCL is bounded by B^2 if $K \ll B$. In most real-life applications of natural language processing, the batch size is usually small considering the large size of existing pretrained language models (e.g., 16,

32, and 64 are common batch sizes, and a maximum of 32 is chosen for experiments in Section IX), the computational cost of FoCL is not high.

VIII. SIMULATION

In this section, we conduct experiments using simulated datasets to validate our findings and our proposed method.

A. Comparison of CE and FoCL

We generate a simulated binary classification dataset $\{x_i, y_i\}_{i=1, \dots, I}$, where $I = 5,000$ and $y_i \in \{0, 1\}$, using the scikit-learn library.¹ The data points are generated as 8 clusters of normally distributed (standard deviation of 1) about vertices of a 128-dimensional hypercube with sides of length of 1. Each data point x_i is 128-dimensional. Each cluster has the same number of data points and is assigned with the same class label.

We use a multi-layer perceptron (MLP) as the classification model: a linear projection layer followed by a Sigmoid activation layer, projecting input data points to embeddings with 64 dimensions. Then a linear classifier is used to predict label of data points.

To train the classification model, we split the whole dataset into train, development, and test, with a ratio of 7:1:2. We train the model with Adam optimizer for 1 K epochs, setting batch size as 1 K and learning rate as 0.01. We compare the trained embeddings under CE loss, and a combination of CE and FoCL where $\lambda = 0.1$. For FoCL, the hyperparameter temperature τ is 0.03. We generate 10 pseudo negative instances with $\alpha = 0.1$.

The visualization of the trained embeddings under CE and FoCL is presented in Fig. 1 of Section I. As discussed, with

¹[Online]. Available: <https://scikit-learn.org/stable/index.html>

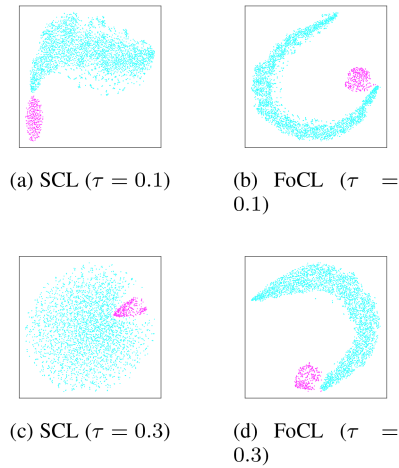


Fig. 7. t -SNE visualization of trained embeddings on a simulated binary classification dataset with high imbalance (9:1). SCL cannot form a dense cluster for the over-represented class (blue) and is sensitive to the temperature hyperparameter τ . FoCL can form dense clusters for both classes.

FoCL, the trained model can form tighter clusters of each class in the latent space. This indicates that FoCL can guide the model to learn more separable features for different classes, mapping instances from different classes far away in the latent space.

B. Comparison of SCL and FoCL

We also conduct simulations to compare the behavior of SCL and FoCL for datasets with high class imbalance. We use the same simulation setup as in Section VIII, but change the number of classes to 2. We keep the ratio of instance number in the two clusters as 9:1. The overall loss function is used as a combination of CE and CL (SCL or FoCL) with $\lambda = 0.3$.

Fig. 7 visualizes the trained embeddings under SCL and FoCL. As shown in the figure, SCL fails to learn a tight cluster for the over-represented class, and the tightness of the over-represented class is sensitive to the hyperparameter temperature τ of the contrastive term. FoCL, on the other hand, learns better separable clusters with decision boundaries that are also easier to learn. This confirms our finding: when a mini-batch is biased towards a certain class, the behavior of SCL will change. For the over-represented class, it will not push instances from the same class together. On the other hand, it will push some of the instances away in the latent space, resulting in a loose cluster.

IX. EXPERIMENTS

We evaluated the proposed method on three benchmark classification tasks: text classification, NER and RE. We also evaluated the proposed method on an end-to-end task which aims to address NER and RE simultaneously.

A. Baselines

CE: Cross-Entropy loss. Minimizes the cross-entropy of the label distributions predicted by the classifier and target label distributions (1).

TABLE III
SUMMARY OF DATASET STATISTICS IN TERMS OF THE NUMBERS OF SENTENCES IN TRAINING, DEVELOPMENT, TEST SET, THE NUMBER OF LABELS

Dataset	#Train	#Dev.	#Test	#Label
<i>Text classification datasets</i>				
EMOTION	3,257	374	1421	4
SEM	4,264	485	878	3
<i>NER datasets</i>				
WNUT16	2,394	1,000	3,849	10
WNUT17	3,394	1,009	1,287	6
CONLL++	14,987	3,466	3,466	4
<i>RE datasets (also used for NER + RE)</i>				
CONLL04	1,153	-	288	4 & 5
SciERC	1,861	275	551	6 & 7
ChEMU	5,911	1,402	2,363	10 & 2

For text classification and NER, we present the number of sentence classes and entity classes, respectively. For RE, we present the number of entity classes and the number of relation classes.

FL: Focal loss [14]. Minimizes the cross-entropy but weighs each instance in the overall loss based on the predicted probabilities (2).

SCL: Contrastive loss combining CE and SCL (4).

FoCL: Proposed *focused* contrastive loss with improved contrastive region, combining CE and FoCL (17).

For CE and FL, we use the framework as shown in Fig. 2(a). For SCL and FoCL, we use the framework as shown in Fig. 2(b).

B. Settings

Our code implementation is based on the HuggingFace² transformers library [44]. We use a batch size of 32 and AdamW optimizer [45] for fine-tuning, where the learning rate is chosen from $\{1e-5, 2e-5, 5e-5, 7e-5\}$. For FL, we set $\gamma = 2$ as suggested by the original paper. For SCL and FoCL, the hyperparameters τ and λ are chosen from $\{0.03, 0.05, 0.07, 0.09, 0.1, 0.3, 0.5, 0.7\}$ and $\{0.1, 0.3, 0.5, 0.7, 0.9\}$, respectively. We find that $\tau = 0.1$ and $\lambda = 0.3$ usually yield the best result for NER and RE models, while text classification models require lower values of temperature. For hard negative mining in FoCL, the number of pseudo negative instances is chosen from $\{32, 48, 64\}$, and we start generating pseudo negative instances after 1/3 of total training steps. For all datasets, we use their standard training, development, and test splits. We evaluate the model performance after each training epoch on the dev. set and select the best checkpoint to compute test results. We run each experiment 5 times with different random seeds and report final average test results.

C. Text Classification

Datasets: We employ two datasets: *EMOTION* which is collected from Twitter [46], and *SEM* which is collected from sustained emotionally coloured machine human interaction [47]. The dataset statistics are summarized in Table III.

²[Online]. Available: <https://huggingface.co/>

TABLE IV

SUMMARY OF TEXT CLASSIFICATION PERFORMANCES IN MICRO-AVERAGED AND MACRO-AVERAGED ACCURACY WHEN VARYING THE NUMBER OF TRAINING INSTANCES

# Sent.	Loss	EMOTION		SEM	
		Micro	Macro	Micro	Macro
50	CE	46.47 $\pm$ 3.17	32.39 $\pm$ 4.18	51.83 $\pm$ 1.91	33.28 $\pm$ 2.56
	FL	46.82 $\pm$ 3.57	33.30 $\pm$ 4.40	50.12 $\pm$ 2.85	34.71 $\pm$ 1.38
	SCL	51.25 $\pm$ 1.14	33.73 $\pm$ 1.79	49.73 $\pm$ 2.82	36.85 $\pm$ 1.46
	FoCL	52.01$\pm$1.83	35.81$\pm$2.52	52.14$\pm$3.36	38.23$\pm$1.96
100	CE	56.28 $\pm$ 3.60	46.68 $\pm$ 3.75	55.31 $\pm$ 2.51	44.51 $\pm$ 3.93
	FL	56.80 $\pm$ 3.65	47.44 $\pm$ 3.75	54.32 $\pm$ 2.55	43.30 $\pm$ 2.40
	SCL	63.52 $\pm$ 1.56	55.94 $\pm$ 1.59	57.02 $\pm$ 1.62	48.27 $\pm$ 4.28
	FoCL	65.08$\pm$0.95	57.57$\pm$1.30	57.52$\pm$2.11	50.19$\pm$3.00
500	CE	74.06 $\pm$ 1.20	69.76 $\pm$ 1.51	61.56 $\pm$ 0.54	57.13 $\pm$ 1.42
	FL	74.40 $\pm$ 1.24	70.29 $\pm$ 1.43	62.07 $\pm$ 1.48	57.32 $\pm$ 1.48
	SCL	75.15 $\pm$ 0.94	70.89 $\pm$ 1.01	62.07 $\pm$ 0.60	58.17 $\pm$ 1.31
	FoCL	76.52$\pm$0.44	72.76$\pm$0.61	62.29$\pm$0.61	58.26$\pm$0.50
Full	CE	79.58 $\pm$ 0.41	75.88 $\pm$ 0.70	62.84 $\pm$ 0.61	59.67 $\pm$ 0.66
	FL	80.14 $\pm$ 0.15	76.90$\pm$0.33	62.45 $\pm$ 0.64	59.00 $\pm$ 1.04
	SCL	80.11 $\pm$ 0.46	76.75 $\pm$ 0.68	62.91 $\pm$ 0.86	59.93 $\pm$ 0.96
	FoCL	80.53$\pm$0.33	76.86 $\pm$ 0.44	63.02$\pm$0.40	60.19$\pm$0.24

For each few-shot scenario (e.g., # sentences = 50), we create a training set by selecting the first sentences (e.g., First 50 sentences) from the original training set.

Model Architecture: We use BERT_{base} [3] to encode all input sentences. The encoded representations are fed into a MLP layer which maps each sentence to a predicted label.

Text Classification Results in Few-Shot Scenarios: Table IV reports the performances of all loss functions in few-shot scenarios, where we report the micro-averaged and macro-averaged accuracy. From Table IV, we see that FoCL obtains significant and consistent improvement over baselines. On EMOTION dataset, FoCL outperforms CE by 5.54, 8.8, and 2.46 in micro-averaged accuracy when the number of training sentences is 50, 100, and 500, respectively. We observe more improvement when the number of training sentences is limited (e.g., 50, 100). This indicates that FoCL is more effective when the supervision signal is limited. Similar results are also observed on SEM dataset, where FoCL obtains more improvement than CE when the number of training sentences is 50.

Text Classification Results on Full Datasets: When full training set is used, FoCL still obtains competitive classification accuracy with more stable performance. On SEM, all loss functions have similar performances. On EMOTION, FoCL still obtains 0.95 and 0.98 improvement over CE in terms of micro-averaged and macro-averaged accuracy. FoCL does not obtain the highest macro-accuracy, however, it is very close to the best performing loss function, FL. In addition, we observe that FoCL in general has less variation (i.e., smaller standard deviation) compared with baselines.

D. Named Entity Recognition

Datasets: We employ two datasets collected from social media, e.g., Twitter and StackExchange, from the WNUT16 [48] and WNUT17 [49] shared tasks. We employ one news datasets CONLL++ [50]. The statistics of all NER datasets are summarized in Table III.

TABLE V

SUMMARY OF NER PERFORMANCES IN MICRO-AVERAGED F1-SCORES WHEN VARYING THE NUMBER OF TRAINING INSTANCES

# Sent.	Loss	WNUT16	WNUT17	CONLL++
50	CE	25.49 $\pm$ 0.34	24.95 $\pm$ 9.26	56.86 $\pm$ 1.31
	FL	23.05 $\pm$ 2.11	24.74 $\pm$ 6.77	53.62 $\pm$ 1.41
	SCL	28.09 $\pm$ 1.07	33.92 $\pm$ 1.67	59.11 $\pm$ 1.11
	FoCL	29.35$\pm$1.66	35.75$\pm$2.07	60.01$\pm$1.28
100	CE	24.05 $\pm$ 8.05	41.92 $\pm$ 1.64	66.56 $\pm$ 0.80
	FL	22.35 $\pm$ 7.02	41.82 $\pm$ 1.39	64.40 $\pm$ 1.95
	SCL	33.54 $\pm$ 5.97	42.81 $\pm$ 2.03	67.02 $\pm$ 0.80
	FoCL	39.69$\pm$1.16	45.97$\pm$2.03	67.94$\pm$0.73
500	CE	49.36 $\pm$ 0.81	50.04 $\pm$ 3.15	77.19 $\pm$ 0.97
	FL	49.67 $\pm$ 0.79	51.18 $\pm$ 0.86	76.97 $\pm$ 0.56
	SCL	50.75 $\pm$ 0.60	50.56 $\pm$ 0.90	78.34 $\pm$ 1.07
	FoCL	51.69$\pm$0.53	53.98$\pm$0.82	79.16$\pm$1.19
Full	CE	52.14 $\pm$ 0.31	56.05 $\pm$ 0.31	92.46 $\pm$ 0.13
	FL	52.69 $\pm$ 0.84	58.80 $\pm$ 0.35	92.08 $\pm$ 0.28
	SCL	53.51 $\pm$ 1.00	58.48 $\pm$ 0.52	92.48 $\pm$ 0.14
	FoCL	55.08$\pm$0.31	59.44$\pm$0.42	92.63$\pm$0.16

Best results are highlighted in bold. For each few-shot scenarios (e.g., # sentences = 50), we create a training set by selecting the first sentences (e.g., first 50 sentences) from the original training set.

Model Architecture: We employ a simple encoder structure where we use a BERT model to encode input sentences with a linear projection layer on top of BERT. We consider different variants of BERT including BERT_{base}, BERT_{tweet}_{base} [6], and BioBERT_{base} [33]. For each dataset, we choose the best performing encoder under CE.

NER Results in Few-Shot Scenarios: Table V reports the performances of all loss functions in few-shot scenarios, where we vary the number of training sentences in {50, 100, 500}. Compared with the scenario where full training data is used, we see that incorporating FoCL is more helpful in few-shot settings when the training data is limited. The improvements of FoCL over baselines are most significant when the training set is smallest, i.e., # sentences = 50. We also find that the re-weighting strategy used by FL is not effective in few-shot scenarios, with performances sometimes even lower than CE. Finally, we observe that FoCL outperforms SCL consistently. This confirms the effectiveness of FoCL and its robustness against data imbalance.

NER Results on Full Dataset: Table V reports the performances of all loss functions on full training set. We see that the incorporation of FoCL achieves the best performances on all datasets consistently. In particular, FoCL achieves more significant gains on more challenging datasets (e.g., WNUT16) where the baselines have poor performances. On WNUT16 and WNUT17, FoCL achieves 5.64% and 6.05% in F1-score compared with CE.

E. Relation Extraction

Datasets: We employ three benchmark English datasets for RE: CONLL04, SciERC, and ChEMU, which are publicly available. Note that we also use the three datasets for joint modelling of NER and RE. All three datasets contain annotated entities and relations between entities, which allow us to train models

TABLE VI
SUMMARY OF NER + RE PERFORMANCES IN MICRO-AVERAGED F1-SCORES WHEN VARYING THE NUMBER OF TRAINING INSTANCES

# Sent.	Loss	CONLL04			SciERC			ChEMU		
		NER	RE	Joint	NER	RE	Joint	NER	RE	Joint
50	CE	60.09 $\pm$ 1.71	24.59 $\pm$ 0.68	23.93 $\pm$ 0.60	44.03 $\pm$ 0.93	15.12 $\pm$ 1.19	7.11 $\pm$ 0.78	82.57 $\pm$ 0.74	77.97 $\pm$ 0.39	69.36 $\pm$ 0.51
	FL	59.73 $\pm$ 1.62	24.71 $\pm$ 0.88	23.93 $\pm$ 0.74	44.25 $\pm$ 0.99	17.07 $\pm$ 1.21	7.98 $\pm$ 0.83	82.01 $\pm$ 0.57	77.84 $\pm$ 0.35	69.02 $\pm$ 0.31
	SCL	60.02 $\pm$ 0.84	23.60 $\pm$ 1.48	23.39 $\pm$ 1.26	43.87 $\pm$ 1.00	17.59 $\pm$ 1.03	8.36 $\pm$ 0.67	82.37 $\pm$ 0.20	78.80 $\pm$ 0.58	69.12 $\pm$ 0.58
	FoCL	63.34 $\pm$ 0.97	26.85 $\pm$ 1.47	26.04 $\pm$ 1.44	45.35 $\pm$ 0.95	18.92 $\pm$ 0.72	9.83 $\pm$ 0.79	82.20 $\pm$ 0.67	78.32 $\pm$ 0.89	68.43 $\pm$ 0.97
100	CE	67.53 $\pm$ 0.97	34.95 $\pm$ 0.82	33.84 $\pm$ 1.15	50.65 $\pm$ 1.24	25.40 $\pm$ 1.08	15.01 $\pm$ 0.88	86.49 $\pm$ 0.23	83.59 $\pm$ 0.57	76.31 $\pm$ 0.53
	FL	67.45 $\pm$ 0.99	35.12 $\pm$ 0.95	34.02 $\pm$ 1.21	50.25 $\pm$ 1.34	23.98 $\pm$ 1.00	12.84 $\pm$ 1.84	86.31 $\pm$ 0.38	83.27 $\pm$ 0.64	76.19 $\pm$ 0.47
	SCL	68.39 $\pm$ 1.22	35.19 $\pm$ 1.11	34.42 $\pm$ 1.19	50.85 $\pm$ 0.95	29.14 $\pm$ 1.28	16.62 $\pm$ 1.45	86.57 $\pm$ 0.45	84.48 $\pm$ 0.64	77.57 $\pm$ 0.57
	FoCL	68.66 $\pm$ 0.99	36.14 $\pm$ 1.00	35.06 $\pm$ 1.18	52.23 $\pm$ 0.87	29.37 $\pm$ 0.92	17.35 $\pm$ 0.84	86.80 $\pm$ 0.51	84.77 $\pm$ 0.77	77.11 $\pm$ 0.87
500	CE	71.73 $\pm$ 0.72	45.79 $\pm$ 0.82	45.55 $\pm$ 0.66	63.57 $\pm$ 0.44	42.77 $\pm$ 1.69	30.08 $\pm$ 1.25	93.11 $\pm$ 0.17	92.76 $\pm$ 0.20	88.43 $\pm$ 0.15
	FL	71.29 $\pm$ 0.74	45.05 $\pm$ 0.87	44.58 $\pm$ 0.65	64.34 $\pm$ 0.42	42.69 $\pm$ 1.59	30.99 $\pm$ 1.33	93.01 $\pm$ 0.21	92.38 $\pm$ 0.26	88.27 $\pm$ 0.28
	SCL	72.54 $\pm$ 0.89	47.55 $\pm$ 0.77	47.32 $\pm$ 0.63	64.67 $\pm$ 0.51	43.23 $\pm$ 0.93	30.58 $\pm$ 0.86	93.09 $\pm$ 0.07	92.92 $\pm$ 0.22	88.39 $\pm$ 0.09
	FoCL	72.35 $\pm$ 0.58	47.66 $\pm$ 0.96	46.98 $\pm$ 0.69	64.56 $\pm$ 0.56	44.01 $\pm$ 0.52	31.06 $\pm$ 0.62	93.34 $\pm$ 0.17	92.90 $\pm$ 0.25	88.69 $\pm$ 0.31
Full	CE	89.15 $\pm$ 0.28	70.81 $\pm$ 0.97	70.99 $\pm$ 1.19	68.64 $\pm$ 0.55	49.75 $\pm$ 0.93	37.54 $\pm$ 1.09	95.53 $\pm$ 0.10	94.99 $\pm$ 0.22	92.62 $\pm$ 0.27
	FL	88.47 $\pm$ 0.27	70.08 $\pm$ 0.35	69.74 $\pm$ 0.78	68.21 $\pm$ 0.64	49.68 $\pm$ 1.02	37.11 $\pm$ 1.29	95.56 $\pm$ 0.09	94.97 $\pm$ 0.25	92.57 $\pm$ 0.19
	SCL	88.66 $\pm$ 0.55	70.26 $\pm$ 0.58	70.32 $\pm$ 0.77	68.71 $\pm$ 0.59	50.71 $\pm$ 1.07	37.88 $\pm$ 1.17	95.67 $\pm$ 0.04	95.09 $\pm$ 0.05	92.63 $\pm$ 0.09
	FoCL	89.13 $\pm$ 0.49	71.45 $\pm$ 1.11	71.36 $\pm$ 1.02	68.88 $\pm$ 0.50	51.17 $\pm$ 1.03	38.79 $\pm$ 1.24	95.77 $\pm$ 0.10	95.31 $\pm$ 0.09	92.80 $\pm$ 0.16

NER: Named entity recognition; RE: Relation extraction assuming entities are known; joint: end-to-end modelling of NER and RE. Best results are highlighted in bold. For each few-shot scenario (e.g., # sentences = 50), we create a training set by selecting the first sentences (e.g., first 50 sentences) from the original training set.

for joint NER and RE. Specifically, CONLL04 [51] contains sentences with annotated named entities and relations extracted from news articles. SciERC [52] is derived from 500 abstracts of artificial intelligence papers. ChEMU [53], [54] contains 1,500 snippets from chemical patents with annotated chemicals and the relationships between chemicals. The statistics of all NER + RE datasets are summarized in Table III.

Model Architecture: We employ the state-of-the-art architecture, SpERT [55], as the encoder in our framework (Fig. 2) for joint modelling of NER and RE. SpERT is a span-based model which uses BERT as its core and two classifiers, span classifier and relation classifier for NER and RE, respectively. The BERT parameters and the two classifiers can be finetuned jointly to address two tasks simultaneously. For conciseness of presentation, we do not detail the structure of SpERT in this paper. In training, we incorporate the FoCL (and other baselines) into the loss function. For all hyperparameters in SpERT that are not related to loss functions, we follow the setting as reported in [55].

RE and NER + RE Results in Few-Shot Scenarios: Table VI reports the performances of all loss functions in few-shot scenarios, where we vary the number of sentences in training set in {50, 100, 500}. Here, RE represents the task of relation extraction only which assumes that are entities are already known, while joint NER + RE is an end-to-end task which performs NER and RE simultaneously. We observe a similar phenomenon as in Section IX-D: in general, FoCL benefits classification tasks more when the training data is less. FoCL outperforms baselines consistently and significantly on CONLL04 and SciERC when number of sentences is 50 and 100. When the number of sentences increases to 500, FoCL obtains competitive performance with SCL in this particular setting and significant improvements over CE and FL. To given an example, for joint NER + RE on SciERC, FoCL improves over CE by 0.98 in F1-score when the number of training sentences is 500. This improvement is enhanced to 2.34 in F1-score when the number of training sentences reduces to 100, and 2.72 when the number of training

TABLE VII
ABLATION STUDY OF FoCL ON TEXT CLASSIFICATION, NER, AND JOINT NER + RE. 500 SENTENCES ARE USED FOR TRAINING

Dataset	CE	Aug.	PM	HN	FoCL
<i>Text Classification</i>					
EMOTION	69.76	69.98	71.54	72.06	72.76
SEM	57.13	57.28	57.76	57.85	58.26
<i>NER</i>					
WNUT16	49.36	49.67	51.05	50.33	51.69
WNUT17	50.04	50.00	53.02	51.66	53.98
CONLL++	77.19	77.20	78.94	78.05	79.16
<i>NER + RE</i>					
CONLL04	45.55	45.75	47.92	46.75	47.99
SciERC	30.08	30.04	30.85	30.71	31.06
ChEMU	88.43	88.40	88.56	88.64	88.69

Aug.: Only with data augmentation (augmented by dropout layer); PM: Positive masking. For text classification, macro-averaged accuracy is reported. For NER and joint NER + RE, micro-averaged F1-score is reported.

sentences is further reduced to 50. On ChEMU dataset, there is less improvement observed for FoCL. We hypothesize that this is due to the high accuracy of baseline classifier trained under CE (See more discussion in Section X).

RE and NER + RE Results on Full Dataset: Table VI reports the performances of all loss functions for NER, RE, and joint NER + RE, respectively. We see that FoCL achieves competitive accuracy on all datasets with some improvement. FoCL achieves more improvement in RE and Joint tasks, which is in align with the few-shot scenarios. In particular, for the RE task, FoCL improves over SCL by 1.19 in F1-score on CONLL04 and 0.46 in F1-score on SciERC. For Joint task, FoCL improves over SCL by 1.04 in F1-score on CONLL04 and 0.91 in F1-score.

F. Ablation Study

Table VII reports the results of ablation study for text classification, NER, and joint NER + RE, where Aug. represents

TABLE VIII
SENSITIVITY TO HYPERPARAMETER α

Dataset	0	0.2	0.4	0.6	0.8
<i>Text Classification</i>					
EMOTION	71.55	71.97	72.76	72.06	71.32
SEM	57.76	58.02	58.26	57.85	57.46
<i>NER</i>					
WNUT16	51.05	51.60	51.69	51.24	51.04
WNUT17	53.04	53.48	53.98	53.19	53.01
CONLL++	78.89	78.99	79.16	78.90	78.80
<i>NER + RE</i>					
CONLL04	47.90	47.90	47.99	47.85	47.81
SciERC	30.88	31.00	31.06	30.98	30.90
ChEMU	88.56	88.67	88.69	88.60	88.50

500 sentences are used for training. For ner and joint NER + RE, micro-averaged F1-score is reported.

adding the data augmentation module ($Aug(\cdot)$ in Fig. 2) to CE, PM represents adding positive masking to Aug., HN represents adding hard negative mixing to Aug., and FoCL represents adding both positive masking and hard negative mixing. We observe a generally consistent increase in performances when more components are added to the framework. This confirms that each component contributes positively to the overall framework. We also observe that in general, PM is a more effective strategy compared with HN. We hypothesize that this is due the limited magnitude of E_m in (20): solely introducing pseudo negative samples is not enough to outweigh the impact of positive instances on an anchor instance in most cases. Note that the data augmentation is performed by a standard dropout layer which does not generate new data instances, and hence, no consistent improvement is found when only using data augmentation with CE. This actually highlights the effectiveness of the proposed FoCL.

Sensitivity to α : The hyperparameter α in (15) used in hard negative mining controls the contribution of the anchor instance in a synthesized hard negative sample. When $\alpha = 0$, the synthesized negative sample has the same embedding as the real hard negative sample. When $\alpha = 1$, the synthesized negative sample has the same embedding as the anchor instance. As mentioned previously, we keep the value of α within (0, 0.4), so that the contribution of the real hard negative sample has higher contribution than the anchor instance. Next, we conduct sensitivity study to α .

Table VIII reports the results of all datasets when we vary the value of α from {0, 0.2, 0.4, 0.6, 0.8}. We see that in general, a model performs better when α increases from 0. This is as expected, since when α increases from 0, the pseudo negative samples will move closer to the anchor instance from the real negative instance. This means the pseudo samples will be more on the decision boundary between the anchor instance's class and the negative class. This will force the model to further push the anchor instance from the decision boundary, leading to tighter clusters. After reaching peak performance at $\alpha = 0.4$, the model degrades with any further increase of α . We hypothesize that this is due to the closeness of pseudo negative samples to the anchor instance in such cases. When pseudo negatives are too close to

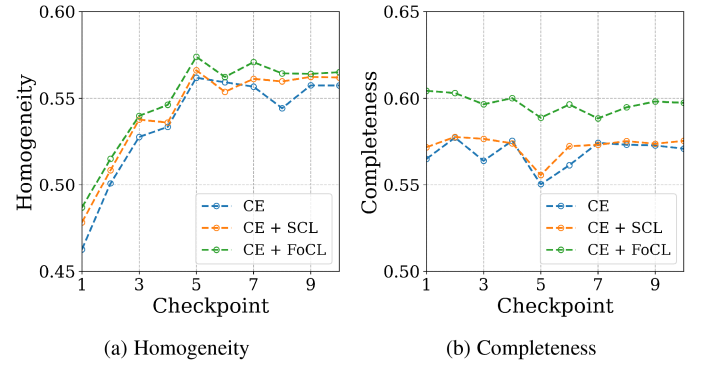


Fig. 8. Cluster homogeneity and completeness of NER models on WNUT16.

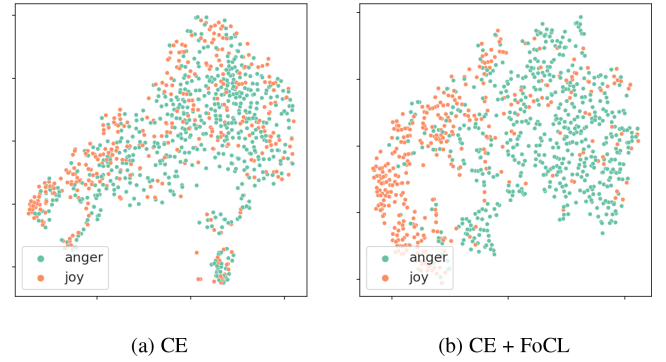


Fig. 9. t -SNE visualization of trained embeddings on EMOTION dataset on class "anger" and "joy" in the few-shot learning setting (# sentences = 50).

the anchor instance, they will lie within the cluster of the anchor instance, and thus, create confusion for the model.

G. Quality of Clusters

We investigate the quality of clusters formed by different models using two metrics: homogeneity and completeness [56]. Specifically, a clustering result satisfies homogeneity if all of its clusters contain only data points that are members of a single class, while it satisfies completeness if all the data points that are members of a given class are elements of the same cluster. For both metrics, higher values indicate better quality of clustering.

We compare the homogeneity and completeness of NER models trained under CE, CE + SCL, and CE + FoCL in Fig. 8, where the homogeneity and completeness of the last 10 checkpoints on WNUT16 are visualized. We see that for both CE and CE + FoCL, homogeneity increases steadily and converges after the 7th checkpoint. CE + FoCL achieves higher homogeneity than CE with slight improvement over CE + SCL. In terms of completeness, CE + FoCL demonstrates consistent and significant improvement compared with both baselines. All models experience moderate fluctuations in the first few checkpoints and converge after the 7th checkpoint, where CE + FoCL maintains consistently higher completeness in every checkpoint than baselines. In particular, CE + FoCL converges at approximately 0.60 in completeness, while CE converges at 0.56. These confirm that with FoCL, models are able to

learn better clustered embeddings via directly optimizing the distribution of each class.

In Fig. 9, we present the t -SNE visualization of the EMOTION dataset in the few-shot setting, where the number of sentences used for training is 50. The figure shows clear difference between CE and CE + FoCL in the trained clusters. For CE, the clusters of “anger” and “joy” are significantly overlapped, whereas the two clusters are better separated when FoCL is introduced.

X. DISCUSSION ON THE USAGE OF FOCL

Both the theoretical analysis and experimental results from above sections confirm the merits of the proposed FoCL loss. Yet, there are some caveats on the usage of FoCL.

FoCL is More Effective for Classification Tasks With a Small Number of Classes and High Class Imbalance: In literature, the main benefit of using contrastive loss for classification tasks is considered as to facilitate a model to form tighter clusters. However, as the theoretical analysis in Section IV-B shows, the aforementioned benefit will not hold for SCL in cases where each training sample has multiple positive instances (instances from the same class as the training sample). Such cases are common, especially when there is high class imbalance: if a training sample is from the majority class, then it is highly likely that it has many positive instances within a training batch. Furthermore, considering the common batch sizes used in NLP tasks, e.g., 8, 16, and 32, the benefits of using FoCL will be more significant when a task has smaller number of classes. For classification tasks with smaller number of classes, it is more likely that within a sampled batch, there are multiple samples from the same class. On the other hand, if a classification task has a large number of classes (e.g., over 100 classes), it is more likely that samples within a mini-batch all come from different classes. In such cases, after data augmentation, each training sample within a training batch will only have one positive instance, where FoCL and SCL will be effectively the same.

However, we do note that for classification tasks with a large number of classes, if larger batch sizes can be used in training, we expect that FoCL will still be more effective than SCL. Ultimately, the benefit of FoCL over SCL will hold true as long as mini-batches are biased towards one or more classes, given our gradient analysis in Section IV-B. To allow larger batch size to be implemented and using FoCL on large classification tasks, a naive solution is simply to use more powerful hardware (e.g., GPU with larger memory). There are also methodological developments that can be done to increase the batch size, such as using Momentum Contrast (MoCo) [57] which caches the embeddings of samples in previous mini-batches and using gradient caching technique [58]. We leave these explorations to future work.

FoCL is More Effective When the Supervision Signal is Limited: From experimental results, we observe that FoCL is much more effective when the baseline CE performs worse. FoCL provides significant improvements when CE has low F1-scores (e.g., F1-scores under 50), while the improvement of FoCL saturates when CE achieves higher F1-scores. For example, on ChEMU datasets where CE achieves at least 82 in F1-score in

NER task, the improvements of FoCL are marginal. This is as expected. In CE, only the labels are used as supervision signal, while in SCL and FoCL, the model training is not only guided by labels, but also the quality of clusters (e.g., whether clusters are tight or not) – The quality of clusters acts as additional support to the supervise the model training. This additional support is particularly useful when supervision signal from labels is not sufficient, e.g., few-shot learning scenarios and noisy training data. When the labels provide enough supervision to train a good classifier, the superiority of using contrastive loss is less significant. This finding is in line with observations from [16] and [59].

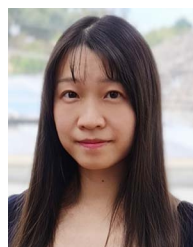
XI. CONCLUSION

In this paper, we studied the incorporation of contrastive learning into classification tasks. We have shown that the vanilla SCL is suboptimal for classification tasks with class distribution variations via gradient analysis. Based on this finding, we have proposed FoCL, which defines a finer contrastive region and improves over SCL in its robustness against variation in class distributions. We have conducted extensive experiments on NLP classification tasks including text classification, NER, RE, and joint modelling of NER and RE. The experimental results confirm that FoCL can improve the prediction accuracy consistently and significantly, especially under few-shot scenarios where training data is limited.

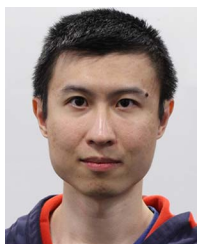
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